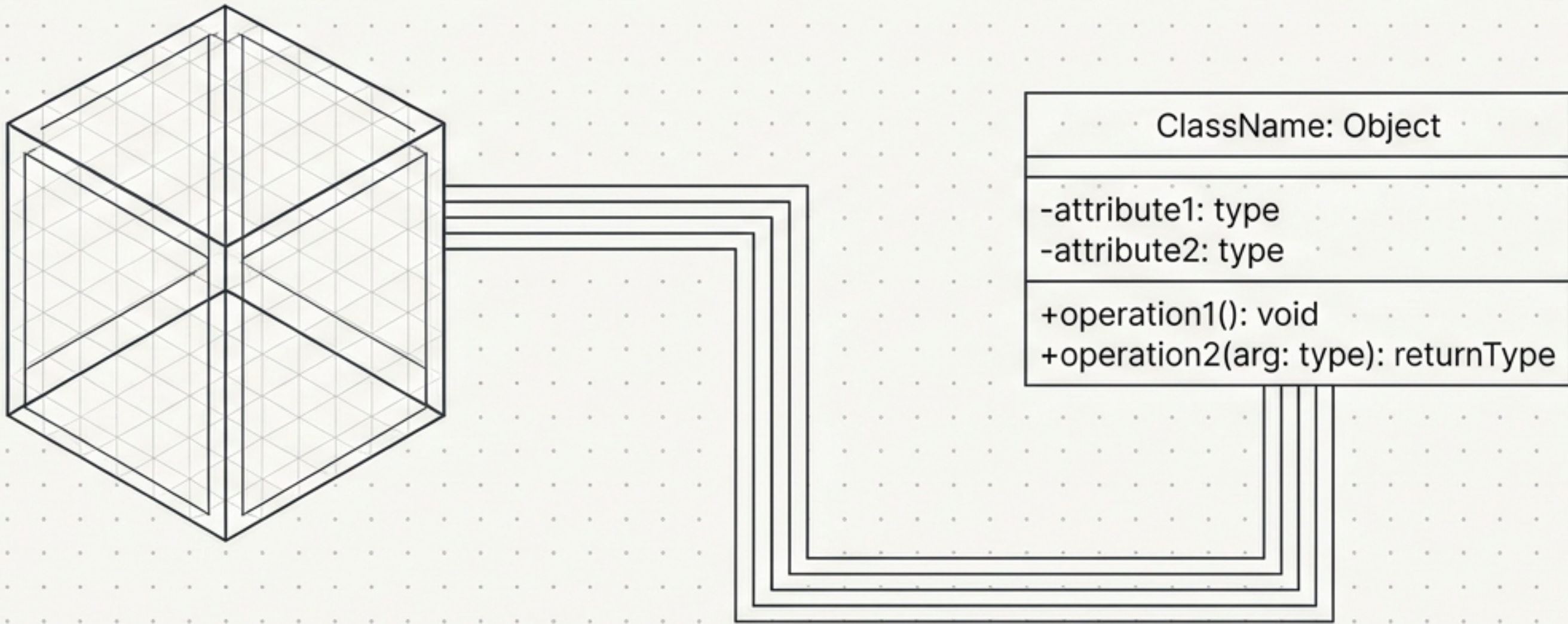
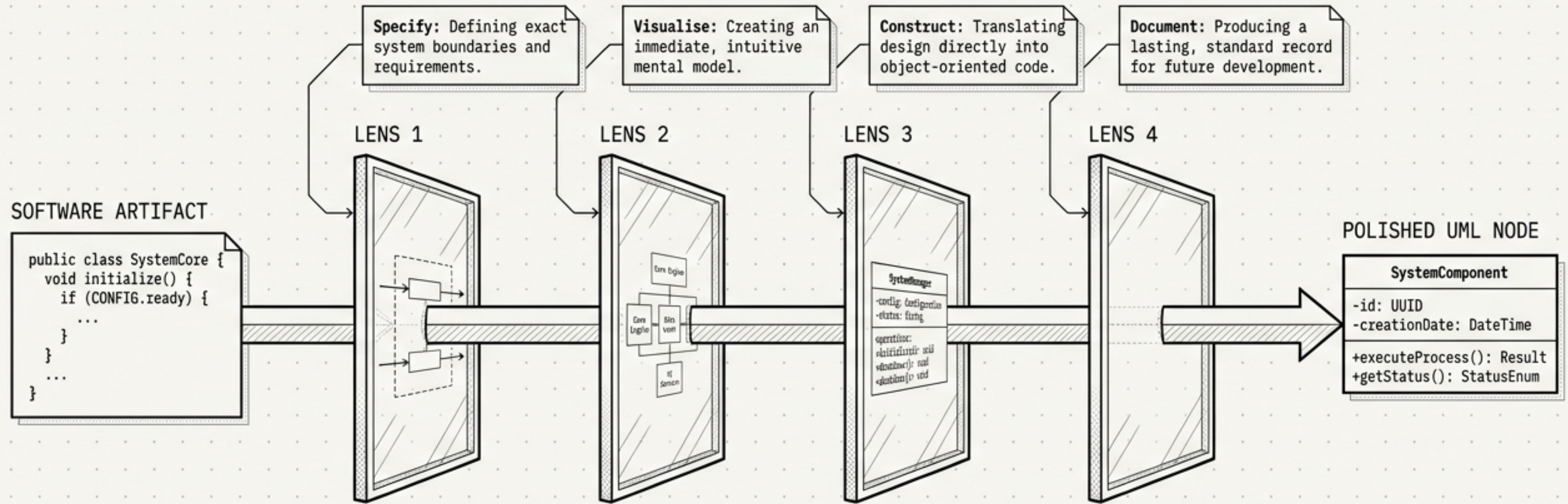


The Anatomy of UML

A visual reference guide for object-oriented system architecture.

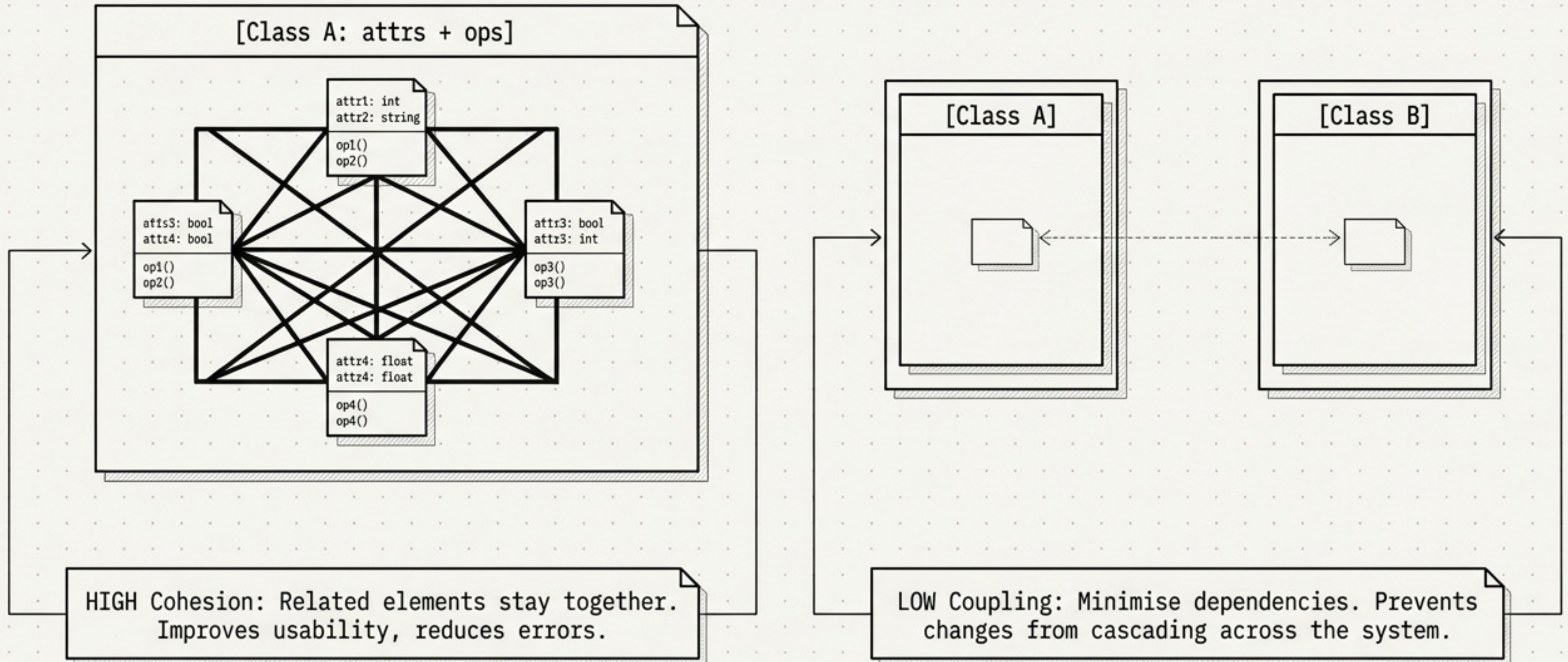


A universal visual language for software-intensive systems

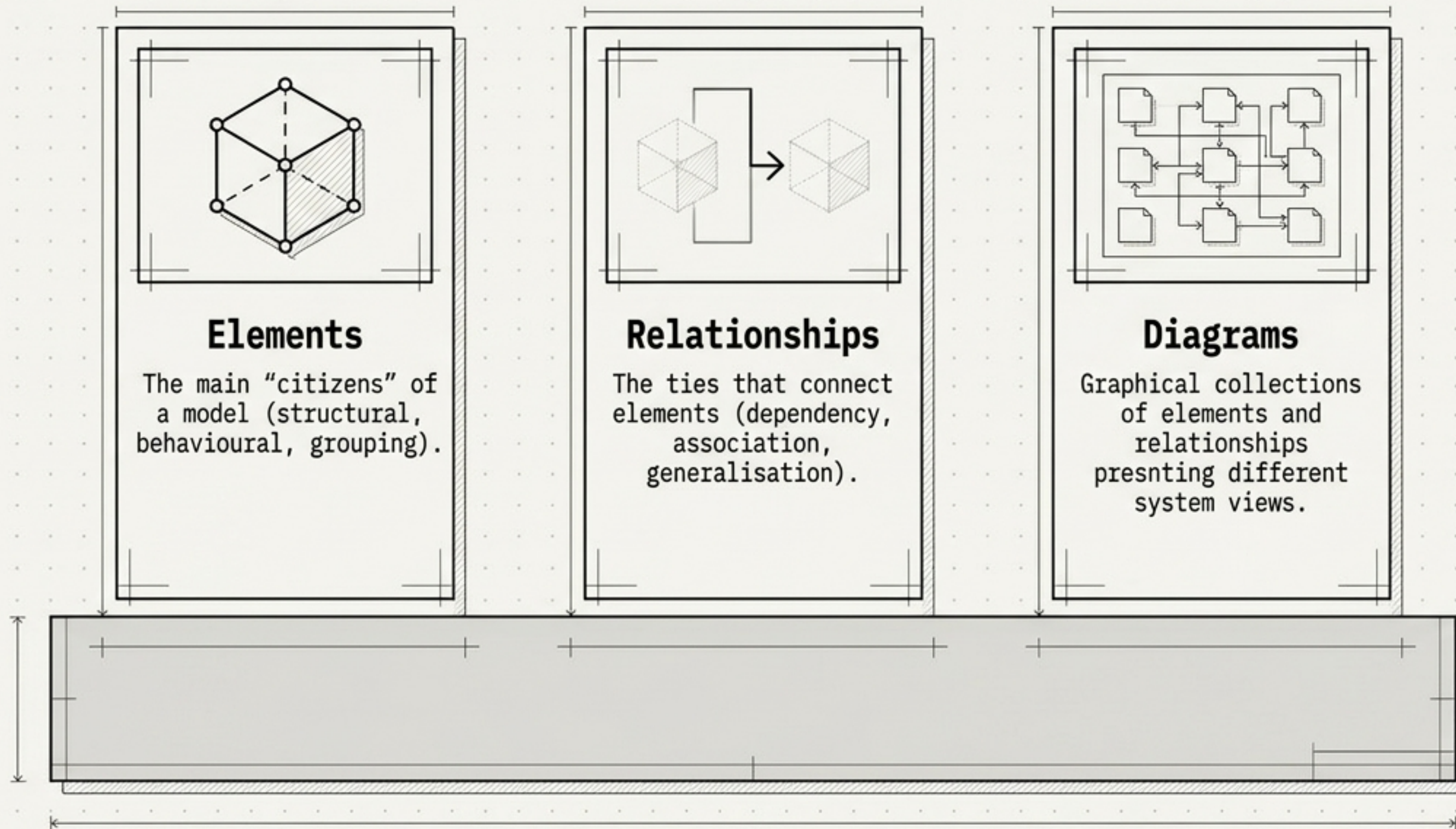


Note that UML provides the standard notation used throughout object-oriented analysis and design.

The physics of a well-designed class



The three fundamental building blocks



Categorising elements: What the system is vs. what it does

STATIC

Structural Elements: Nouns.

Describes what the system is
(e.g., classes, components, nodes).
Does not change at run-time.

Grouping Elements: Organisational.

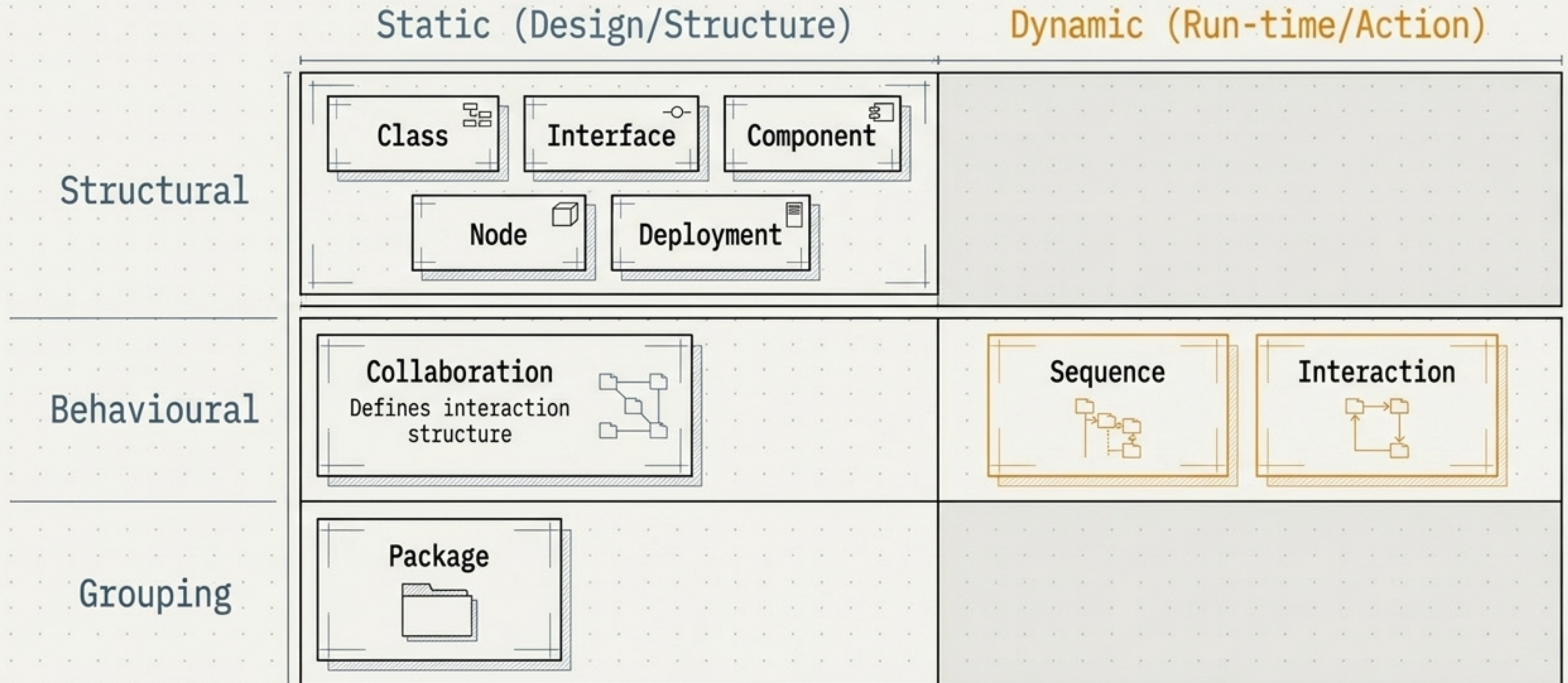
Organises the model (packages).
Purely conceptual, development-time only.

DYNAMIC

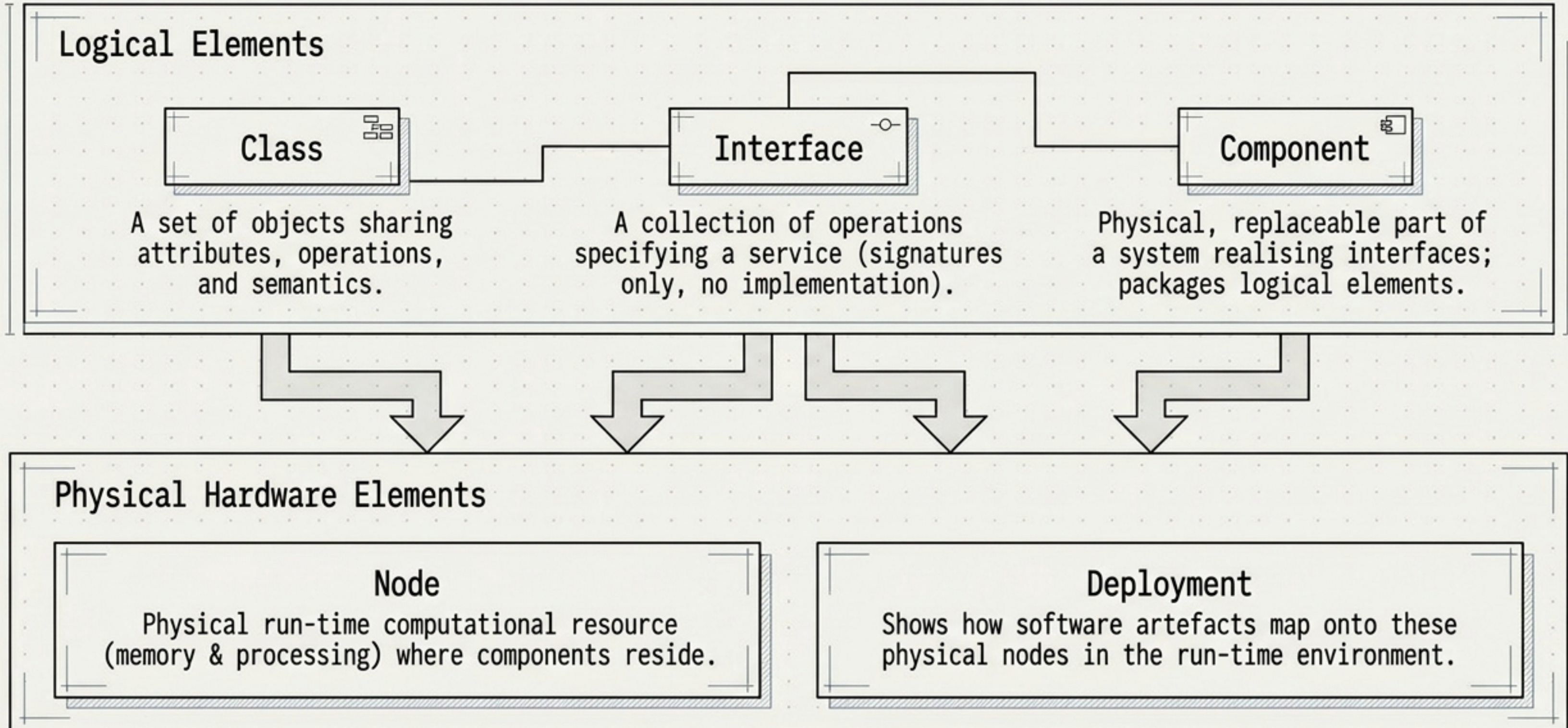
Behavioural Elements: Verbs.

Describes what the system does
(e.g., interactions).
Changes over time.

The Element Ontology Matrix



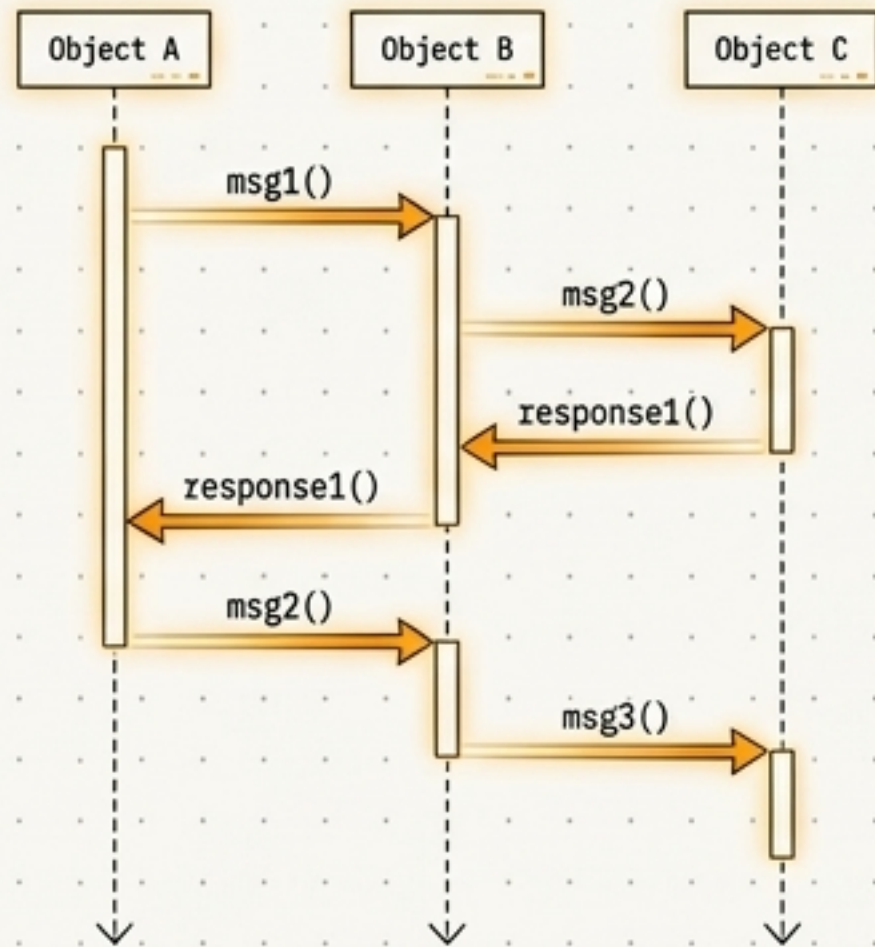
Structural elements map both logical code and physical hardware



Behavioural elements capture chronological and structural interactions

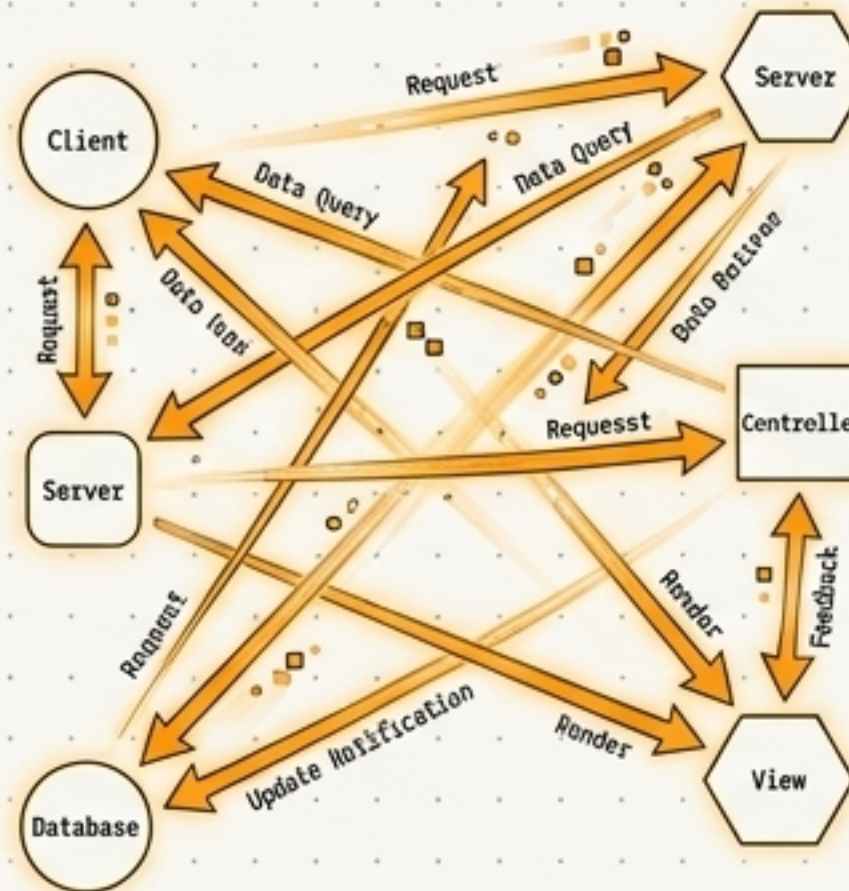
Sequence

Emphasises time. Shows interactions as a set of objects and the messages exchanged between them, strictly ordered chronologically.



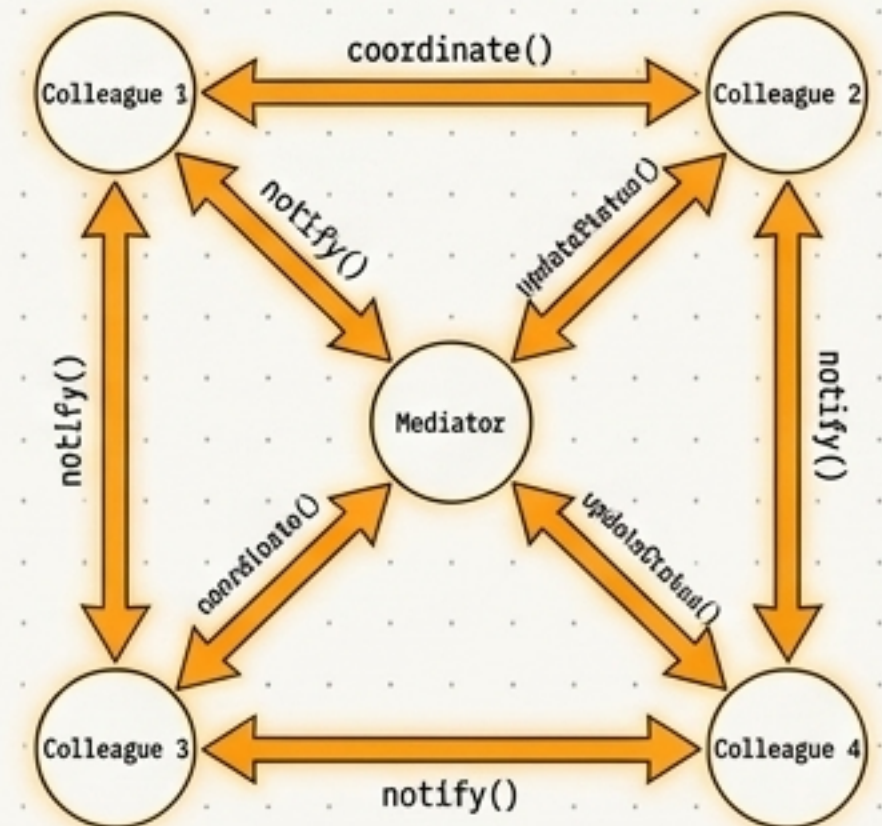
Interaction

The overarching concept. Describes communication among objects (messages, action sequences, links) to achieve a specific purpose.



Collaboration

Emphasises structure. Defines an interaction of cooperating elements that together deliver a behaviour (frequently used to model design patterns).



Classified as both structural and behavioural.

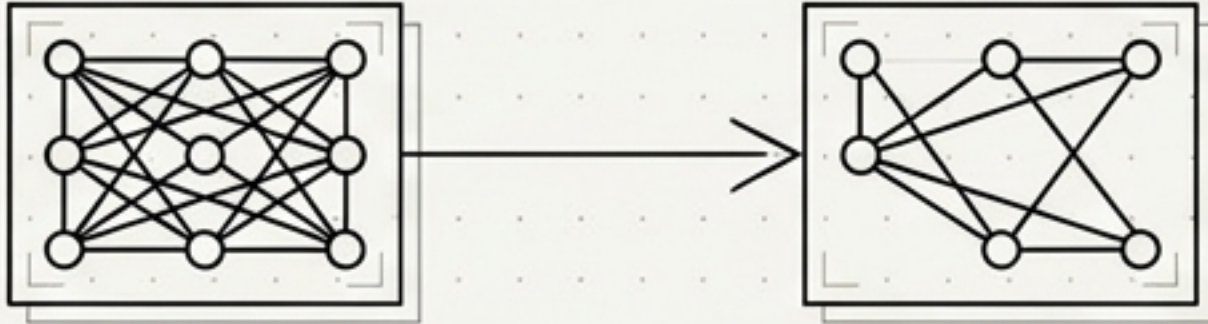
The ties that bind:

A visual dictionary of relationships

Name	Notation	Description
Dependency		Semantic relationship. A change to the independent element affects the dependent element.
Association		Structural relationship describing a set of links between objects. (Aggregation is a whole-part form).
Generalisation		Inheritance ("is-a"). Objects of the specialised element can substitute the generalised element.
Realisation		Contract + Guarantee. One element specifies a contract (interface), another guarantees to carry it out.

The UML Master Dashboard

Core Principles

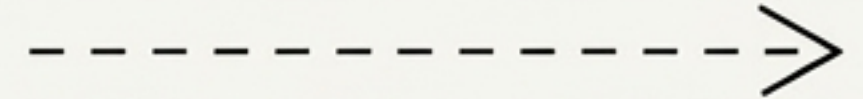


High Cohesion - Inside

Low Coupling - Between

Notation Reference

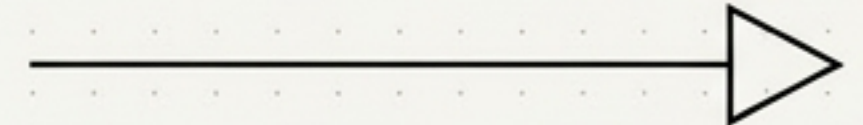
Dependency



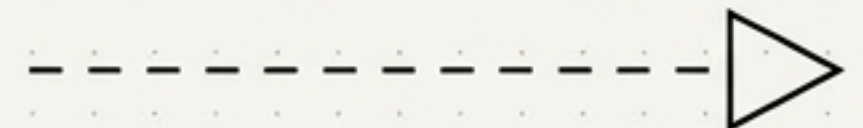
Association



Generalisation



Realisation



Element Taxonomy

Static

(Structural Nouns like
Class, Node, Component)

Dynamic

(Behavioural Verbs like
Sequence, Interaction)

Grouping

(Package)

UML: Specifying, visualising, constructing, and documenting software artefacts.